NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

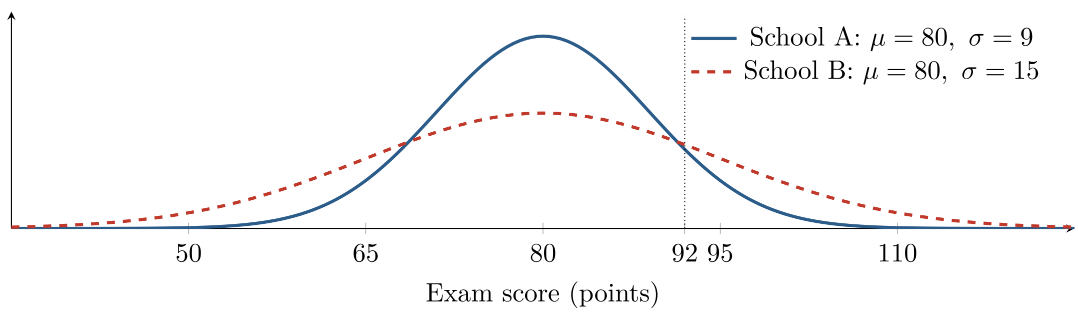
Same Score, Different Standing

AP Statistics · Topic 5.2 (CED 2.11) · B: 2026-11-10 · E: 2026-12-04

[STAR] TODAY'S PROBLEM

Suppose exam scores at School A and School B are each approximately normal. Both distributions have mean 80 points, but School A has standard deviation 9 points and School B has standard deviation 15 points.

A counselor says, "A score of 92 is at about the 90th percentile at both schools because both schools have the same mean." A data coach says, "The different spreads could put 92 at different percentiles."



FIRST TAKE — before the video (5 min)

Is a score of 92 at the same percentile in both schools? Choose a claim and name one parameter behind your choice.

[BOOK] STANDARDIZE BEFORE COMPARING (keep on the page)

For a normal random variable, $z = \dfrac{x-\mu}{\sigma}$ tells how many standard deviations x is from its mean. The percentile of x is the area under the normal curve to the **left** of x, written $P(X \le x) = \Phi(z)$. Raw distances from the mean are comparable only when the standard deviations match. With the same above-mean raw distance, a smaller standard deviation produces a larger positive z-score and a higher percentile.

Finish after the video:

DOK 1 **(a)** Identify which school has the more variable score distribution and state how the corresponding normal curve shows that difference.

DOK 2 **(b)** For a score of 92, calculate the z-score and percentile at each school. Show the standardization and report each percentile to the nearest tenth of a percent.

DOK 3 **(c)** Decide whether the counselor's same-percentile claim is warranted. Cite both z-scores and percentiles, explain why equal means do not give equal relative standing when standard deviations differ, and state at which school a 92 is 'better' if better means a higher standing relative to that school's scores. Explain what the counselor's claim would mislead a reader into believing.

Frame: At School A, $z =$ __________ and the percentile is __________; at School B, $z =$ __________ and the percentile is __________.

Frame: The counselor's claim is __________ because the same 12-point gap is ____________________; therefore a 92 has the higher relative standing at __________.

EXIT — turn this in

One thing the video changed about my first take: ______________________________